Q1: Write an 8086 assembly language program to configure an 8255 PPI such that Port A works as an output port and Port B works as an input port. Read the data from Port B and display it on Port A. Mention the control word used.

Introduction

The Intel 8255 Programmable Peripheral Interface (PPI) is used to interface input/output devices with the 8086 microprocessor. In this program, Port B is configured as an input port to read data (e.g., DIP switches), and Port A is configured as an output port to display the input data (e.g., LEDs). The 8255 operates in Mode 0, which is basic input/output mode.

Port Address

Port	Address
Port A	60H
Port B	61H
Control Register	63H

Control Word Calculation

Control word format-

Bit	Description	Value
D7	Mode set flag	1
D6–D5	Group A mode	00 (Mode 0)
D4	Port A	0 (Output)
D3	Port C upper	0 (Output)
D2	Group B mode	0 (Mode 0)
D1	Port B	1 (Input)
D0	Port C lower	0 (Output)

Control Word = 1000 0010B = 82H

8086 Assembly Language Program

```
PORTA   EQU 60H
PORTB   EQU 61H
CONTROL EQU 63H

        MOV AL, 82H       ;  Port A as output, Port B as input
        OUT CONTROL, AL

MAIN:   IN AL, PORTB      ; Read data from Port B
        OUT PORTA, AL     ; Display data on Port A
        JMP MAIN          ; Continuous operation
```

Conclusion

The given 8086 assembly language program successfully demonstrates the interfacing of the 8255 PPI with the 8086 microprocessor. By using an appropriate control word, Port B was configured as an input port and Port A as an output port in Mode 0 operation.

The program continuously reads the data from Port B and displays it on Port A, validating the correct configuration and functionality of the PPI. This experiment highlights the importance of control word formation and basic input/output operations in microprocessor-based system design. Such interfacing techniques form the foundation for more advanced embedded applications involving sensors, displays, and real-time control systems.

Q2: Explain how ARM Cortex features help in designing a low-power, real-time sensor-based monitoring system.

Introduction

Modern embedded systems often require efficient processing, real-time responsiveness, and minimal power consumption, especially in applications such as sensor-based monitoring systems. ARM Cortex processors are specifically designed to meet these requirements by combining high computational performance with low power usage. Due to their scalable architecture and optimized instruction execution.

ARM Cortex cores are widely used in Internet of Things (IoT), medical devices, industrial automation, and portable electronic systems. This discussion focuses on how key features of ARM Cortex processors—namely interrupt handling, low-power operating modes, and pipeline architecture—contribute to the design of reliable, energy-efficient, and real-time sensor-based monitoring systems.

1. Interrupt Handling

ARM Cortex processors support fast and efficient interrupt handling using features such as Nested Vectored Interrupt Controller (NVIC).
This allows the processor to respond immediately to sensor events without continuous polling. As a result:

- Response time is reduced
- CPU power consumption is minimized
- Real-time performance is improved

2. Low-Power Modes

ARM Cortex processors provide multiple low-power operating modes such as Sleep, Deep Sleep, and Standby modes.
In sensor-based systems, the processor remains in sleep mode most of the time and wakes up only when a sensor interrupt occurs. This significantly:

- Reduces energy consumption
- Extends battery life
- Improves system efficiency

3. Pipeline Architecture

ARM Cortex uses a pipeline architecture where instruction fetch, decode, and execution occur simultaneously.
This enables:

- Faster instruction execution
- Higher throughput
- Real-time processing of sensor data with minimal delay

Conclusion

By combining fast interrupt handling, efficient low-power modes, and pipelined execution, ARM Cortex processors provide an ideal platform for designing low-power, real-time, sensor-based monitoring systems.